SEMESTER – V

SUBJECT NAME: PHP WITH MYSQL

SUBJECT CODE: BCA24-CC-501

CREDIT: 4

CONTACT HOURS: 48

COURSE OBJECTIVE:

1. To understand the fundamentals of PHP programming
2. To learn to design and manage relational databases using MySQL
3. To develop skills to integrate PHP with MySQL
4. To implement form handling and server-side validation
5. To design and deploy database-driven web applications using PHP and MySQL

COURSE OUTCOME:

After successful completion of this course, students will be able to:

CO1: Explain the basic concepts of server-side scripting and write simple programs using PHP syntax and control structures.

CO2: Design and create relational databases using MySQL, including tables, keys, and basic SQL queries.

CO3: Develop PHP programs to connect with MySQL databases and perform CRUD (Create, Read, Update, Delete) operations.

CO4: Implement form handling, session management, and basic security techniques in PHP-based web applications.

CO5: Design and develop a complete dynamic web application using PHP and MySQL for data storage and retrieval.

DETAILED SYLLABUS:

MODULE NO:	NAME OF THE TOPIC	HOURS
M1	**Introduction to Web Development:** Overview of web development, server vs. client-side, HTML basics, introduction to PHP and MySQL Setting up Environment: Installation of XAMPP/WAMP,	8

	introduction to Apache, MySQL, and PHP. Configuring a development environment	
M2	**PHP Basics:** PHP syntax, variables, data types, operators, and control structures. Functions in PHP: Defining and using functions, function scope, including files, and built-in functions Arrays and Strings: Working with arrays, array functions, string manipulation, and string functions	10
M3	**Forms Handling and User Input:** Handling form data, GET vs. POST methods, validating and sanitizing user input **PHP and MySQL Integration:** Connecting PHP to MySQL, executing queries, retrieving data, and displaying data in web pages Implementing Create, Read, Update, Delete (CRUD) operations using PHP and MySQL	10
M4	**Session Management:** Understanding sessions and cookies, creating and managing sessions in PHP. **User authentication:** Building User Login system , Password Hashing. Error Handling andFile Handling In PHP: Reading, Writing and uploading files. Security in PHP: SQL injection prevention, XSS protection, securing file uploads. **Advanced PHP Concepts:** Object-Oriented PHP, classes, objects, inheritance, and PHP's built-in OOP features	10
M5	**Introduction to PHP Frameworks:** Overview of popular PHP frameworks (e.g., Laravel, CodeIAECer), basic MVC architecture concepts **Laravel FrameWork:** Introduction to Laravel, Understanding routing in Laravel, creating and using controllers, passing data to views, and route parameters. Introduction to Blade templating, using Blade directives, template	10

	inheritance, and displaying data in views. Overview of Eloquent ORM, defining models, basic CRUD operations, database migrations, and relationships.	

SUGGESTED READING:

1. Gilmore, W. J. (2010). Beginning PHP and MySQL: From Novice to Professional (4th ed.). Apress.
2. Naramore, E., Gerner, J., Scouarnec, Y. L., Stolz, J., & Glass, M. (2005). Beginning PHP5, Apache, and MySQL Web Development. Wiley India Pvt. Ltd.
3. Welling, L., & Thomson, L. (2009). PHP and MySQL Web Development (4th ed.). Pearson Education.
4. Beighley, L., & Morrison, M. (2008). Head First PHP & MySQL. O'Reilly Media.
5. Powers, D., & McGrath, M. (2015). PHP & MySQL in Easy Steps. McGraw Hill Education.
6. Robson, E., & Freeman, E. (2007). Head First PHP & MySQL. O'Reilly Media.
7. Gosselin, D., Kokoska, R., & Easterbrooks, P. (2010). PHP Programming with MySQL: The Web Technologies Series (2nd ed.). Cengage Learning.
8. Sharma, V. (2017). PHP with MySQL: A Practical Approach. BPB Publications.
9. Narayan, N. (2016). PHP & MySQL: Server-side Web Development. McGraw Hill Education. Zandstra, M. (2014). PHP Objects, Patterns, and Practice (4th ed.). Apress.

SUBJECT NAME: Object Oriented Programming with Java
SUBJECT CODE: BCA24-CC-502
Credit: 4
CONTACT HOURS : 48

COURSE OBJECTIVE:

1. **To introduce the fundamental concepts of Object-Oriented Programming,**
2. **To develop programming skills using Java syntax and constructs, data types, operators, control structures, arrays, and methods.**
3. **To design and implement object-oriented solutions using classes and objects**
4. **To learn advanced Java features, like - packages, interfaces, exception handling, multithreading etc.**
5. **To develop the ability to build modular and reusable programs using object-oriented design principles.**

COURSE OUTCOME

CO1: Explain the fundamental concepts of Object-Oriented Programming such as classes, objects, inheritance, polymorphism, abstraction, and encapsulation using Java.

CO2: Develop Java programs using basic programming constructs, control structures, arrays, and methods.

CO3: Apply object-oriented principles in Java to design reusable and modular programs using classes and objects.

CO4: Implement advanced OOP features such as inheritance, interfaces, packages, exception handling, and multithreading in Java programs.

CO5: Design and develop small-scale applications using Java by applying object-oriented design principles.

DETAILED SYLLABUS:

Module	NAME OF THE TOPIC	Hours

No:		
M1	**Introduction to Java:** Overview of Java, its features, setting up the environment, and writing first Java program. PATH AND CLASS PATH VARIABLE, JVM architecture. **Basics of Java Programming:** Data types, variables, operators, control structures, and loops. Storing of Big Integer. Use of Static variable, final variable, initialization of static and final variable.	10
M2	**Object-Oriented Programming Concepts:** Access modifiers, encapsulation, and abstraction in Java. Classes, objects, methods, and different types of initialization. Calling a constructor from a constructor, constructor chaining. Inheritance and Polymorphism: Inheritance, method overloading, method overriding, and polymorphism. Restriction on method overriding.	10
M3	**Arrays and Strings:** Single-dimensional and multi-dimensional array. Use of Arrays class for printing, sorting, searching. string handling, and use of StringBuffer and StringBuilder. **Interfaces and Abstract Classes:** Understanding interfaces, abstract classes, and multiple inheritance in Java.	10
M4	**Exception Handling:** Checked and Unchecked Exception, Try-catch blocks, multiple catch blocks, finally, throw, throws, and custom exceptions. **File I/O and Serialization:** File handling, reading and writing to files, and serialization in Java.	10
M5	Multithreading: Basics of threads, thread lifecycle, synchronization, and inter-thread communication.	8

SUGGESTED READING:

1. Schildt, H. (2018). Java: The Complete Reference (11th ed.). McGraw Hill Education.
2. Horstmann, C. S., & Cornell, G. (2019). Core Java Volume I – Fundamentals (11th ed.). Pearson Education.

3. Balagurusamy, E. (2020). Object-Oriented Programming with Java (7th ed.). McGraw Hill Education.
4. Sierra, K., & Bates, B. (2005). Head First Java (2nd ed.). O'Reilly Media.
5. Deitel, P., & Deitel, H. (2018). Java: How to Program (11th ed.). Pearson Education.
6. Liang, Y. D. (2018). Introduction to Java Programming and Data Structures (11th ed.). Pearson Education.
7. Halterman, R. L. (2005). Object-Oriented Programming in Java. WCB/McGraw Hill.
8. Tanweer Alam (2021). Core JAVA. Khanna Publishing House.
9. Tanweer Alam (2020). Object Oriented Systems with Java 2.0. Khanna Publishing House.
10. Arunesh Goyal (2020). The Essentials of Java. Khanna Publishing House.

SUBJECT NAME: Entrepreneurship

SUBJECT CODE: BCA24-DSE-503

Credit: 2

CONTACT HOURS : 24

COURSE OBJECTIVE:

1. **To introduce the concept and importance of entrepreneurship and its role in economic and social development.**
2. **To develop entrepreneurial skills and mindset required for identifying business opportunities and taking calculated risks.**
3. **To familiarize students with the process of starting and managing a new venture, including idea generation, planning, and resource management.**
4. **To provide knowledge about business planning, financing, and marketing strategies required for successful entrepreneurship.**
5. **To encourage innovation, creativity, and leadership abilities necessary for becoming successful entrepreneurs.**

Course Outcome:

After successful completion of this course, students will be able to:

CO1: Explain the concept, characteristics, and importance of entrepreneurship in economic development.

CO2: Identify business opportunities and evaluate the feasibility of entrepreneurial ideas.

CO3: Prepare a basic business plan including marketing, financial, and operational aspects.

CO4: Analyze the challenges and risks involved in starting and managing a new venture.

CO5: Demonstrate entrepreneurial skills such as creativity, innovation, leadership, and decision-making in developing business ideas.

MODUL	COURSE CONTENT	Hours

E NO.		
1	**Introduction to Entrepreneurship:** Meaning and concept of entrepreneurship, The history of entrepreneurship development, Factors influencing entrepreneurship, Theories of Entrepreneurship, Role and Importance of Entrepreneurship in Economic Growth, New generations of entrepreneurship viz. social entrepreneurship, Health entrepreneurship, Tourism entrepreneurship, Women entrepreneurship etc., Creativity and entrepreneurship, Steps in Creativity, Barriers to entrepreneurship	4
2	**Introduction to entrepreneur:** Meaning and concept of entrepreneur, types of entrepreneur,Characteristics of Entrepreneurs, Functions of entrepreneur	4
3	**Entrepreneurial Motivation:** Entrepreneurial Motivation, Need for Achievement Theory, Maslow's theory, Herjburg's theory, McGragor's Theory, Risk-taking Behavior, Innovation and Entrepreneur	6
4	**Project Management:** Ideas – Sources, processing; Input Requirements, Sources of Financing, Technical Assistance, Marketing Assistance, Preparation of Feasibility Reports, Legal Formalities and Documentation.	6
5	**Organisation Assistance :** Assistance to an entrepreneur, New Ventures Industrial Park (Meaning, features, & examples), Special Economic Zone (Meaning, features & examples),Financial assistance by different agencies MSME, The Small Industries Development Bank of India(SIDBI),The State Small Industries Development Corporation(SSIDC)	4

Suggested Readings:

1. Entrepreneurship – Anindita Sarkar – Taurean Publications
2. Entrepreneurship Development and Project Management – Dr. Dilip M. Sarwate – Everest Publishing House

SUBJECT NAME: Financial management

SUBJECT CODE: BCA24-DSE-504

Credit: 2

CONTACT HOURS : 24

COURSE OBJECTIVE:

1. **To introduce the basic concepts and scope of financial management and its importance in business decision-making.**
2. **To develop an understanding of financial planning and control.**
3. **To learn investment decision techniques, such as capital budgeting methods.**
4. **To gain knowledge about financing decisions, sources of finance and cost of capital.**
5. **To understand working capital management and the management of short-term financial resources in an organization.**

Course Outcome:

After successful completion of this course, students will be able to:

CO1: Explain the principles and functions of financial management in business organizations.

CO2: Analyze financial statements to evaluate the financial performance of a company.

CO3: Apply capital budgeting techniques to evaluate investment proposals.

CO4: Assess different sources of finance and determine the cost of capital for business decisions.

CO5: Evaluate working capital requirements and manage short-term financial resources effectively.

MODULE NO.	COURSE CONTENT	Hours
1	**Introduction:** Concepts, Nature, Scope, Function and Objectives of Financial Management. Basic Financial Decisions: Investment, Financing and Dividend Decisions. Financial goals - Profit vs. Wealth Maximization; Finance Functions – Investment, Financing and Dividend Decisions – Cost of Capital – Significance of Cost of Capital – Calculation of Cost of Debt – Cost of Preference Capital – Cost of Equity Capital (CAPM Model and Gordon's Model) and Cost of Retained Earnings – Combined Cost	6

	of Capital (weighted/Overall).	
2	**Analysis and Interpretation of Corporate Final Accounts:** Understanding the Parameters of health of Business: Liquidity, Profitability, Solvency and Efficiency through learning computation, analysis and interpretation of various tools of financial analysis Preparation of Cash Flow Statement as per Accounting Standard and its Analysis Capital Budgeting – Nature of Investment Decisions – Investment Evaluation criteria – Net Present Value (NPV), Internal Rate of Return (IRR), Profitability Index (PI), Payback Period, Accounting Rate of Return (ARR) – NPV and IRR comparison.	6
3	**Leverage Analysis:** Developing the Concept of Leverage in Finance. Computation and inferences of Degree of Operating Leverage, Financial Leverage and Combined Leverage – Measurement of Leverages – Effects of Operating and Financial Leverage on Profit – Analyzing Alternate Financial Plans - Capital Structure Theories - Traditional approach - M.M. Hypotheses – without Taxes and with Taxes – Net Income Approach (NI) – Net Operating Income Approach (NOI) - Determining capital structure in practice.	4
4	**Investment Decisions:** Analysis of Risk and Uncertainty. Concept and Computation of Time Value of Money, DCF and Non DCF methods of Investment Appraisal. Project selection on the basis of Investment Decisions. Valuating Investment Proposals for Decision Making. Capital Rationing Dividend Policies – Issues in Dividend Decisions – Relevance Theory – Walter's Model – Gordon's Model – Irrelevance Theory – M-M hypothesis - Dividend Policy in Practice – Forms of Dividends – Stability in Dividend Policy – Corporate Dividend Behaviour.	4
5	**Management of Working Capital:** Concepts, components, Determinants, need, Significance and types of Working Capital – Calculating Operating Cycle Period and Estimation of Working Capital Requirements – Financing of Working Capital and norms of Bank Finance – Sources of Working capital – Factoring services–Various committee reports on Bank Finance – Dimensions of Working Capital Management, Computation of Working Capital for a	4

	Company.	

Suggested Readings:

1. Financial Management: Theory and Practice - by Prasanna Chandra
2. Financial Management" by I. M. Pandey
3. Financial Management: Principles and Applications" by S. N. Maheshwari
4. Strategic Financial Management" by Ravi M. Kishore
5. Financial Management: Text, Problems and Cases" by M. Y. Khan and P. K. Jain.

Practical:
SUBJECT NAME: PHP WITH MYSQL LAB
SUBJECT CODE: BCA24-CC-591
Credit: 2
Contact Hours : 48

Course Objective:
The students will learn

1. **To provide hands-on experience in PHP programming for developing dynamic web pages.**
2. **To create and manage MySQL databases for web applications.**
3. **To connect PHP with MySQL and performing database operations.**
4. **To design and implement web forms and handling user input using PHP.**
5. **To develop the ability to build small database-driven web applications using PHP and MySQL.**

Course Outcomes

CO1: Develop basic PHP programs using variables, operators, and control structures.

CO2: Create and manage MySQL databases, tables, and execute SQL queries.

CO3: Implement PHP scripts to connect with MySQL databases and perform CRUD operations.

CO4: Design web forms and implement server-side validation using PHP.

CO5: Develop a simple dynamic web application integrating PHP and MySQL.

MODULE NUMBER	Name of the topic
1	**User Registration and Login System** ☐ Case Study: A website needs a secure system for user registration and login, allowing users to create accounts and access personalized content. Develop a PHP and MySQL-based system where users can register with a username and password, and log in to access their profile page. Implement password hashing for security.
2	**Content Management System (CMS)** ☐

	Case Study: A blog site requires a CMS that allows authors to write, edit, and publish articles. Design a CMS using PHP and MySQL where authors can log in, create new posts, edit existing posts, and delete posts. The system should display posts to the public in a structured format.
3	**Online Voting System** □ Case Study: An organization wants to conduct online elections where members can vote for their preferred candidates. Create a PHP and MySQL-based voting system where users can register, vote for candidates, and view voting results. Ensure that each user can vote only once.
4	**E-commerce Product Catalog** □ Case Study: An online store needs to manage its product catalog, allowing users to browse products by category and view product details. Implement a product catalog system in PHP and MySQL where users can browse products by category, search for specific products, and view detailed descriptions and prices.
5	**Event Management System** □ Case Study: A company needs a system to manage events, including registration, scheduling, and participant lists. Develop a PHP and MySQL-based event management system where users can register for events, view event schedules, and check participant lists.
6	**Forum or Discussion Board** □ Case Study: A community website needs a forum where users can post topics, reply to discussions, and view threads. Build a forum using PHP and MySQL where users can create new discussion threads, reply to existing threads, and manage their posts. Implement user authentication to control access.

Practical:

SUBJECT NAME: Object Oriented Programming with Java Lab

SUBJECT CODE: BCA24-CC-592

Credit : 2

Contact Hours : 48

Course Objective:

The students will learn

1. **To provide practical exposure to Java programming using basic syntax, control structures, and data types.**
2. **To develop programming skills using classes and objects in Java.**
3. **To demonstrate the implementation of object-oriented concepts such as inheritance, polymorphism, encapsulation, and abstraction.**
4. **To enable students to implement advanced Java features such as packages, interfaces, and exception handling.**
5. **To develop the ability to design and implement small applications using object-oriented programming principles in Java.**

MODULE NUMBER	Name of the topic
1	A library wants to automate book issuing and returning processes. The system should track available books and issued books. Create classes Book, Library, and Member. Implement methods to issue a book to a member, return a book, and display the list of available books.
2	A company requires a payroll system to calculate employee salaries based on hours worked and hourly rate. Develop an Employee class with attributes like name, id, hoursWorked, and hourlyRate. Implement methods to calculate the salary and generate a payroll report for all employees.
3	A bank wants to manage customer accounts and transactions, including deposits, withdrawals, and balance inquiries. Create classes Account, Customer, and Transaction. Implement methods to perform deposits, withdrawals, and display account balance. Handle exceptions for insufficient balance during withdrawals.
4	A rental service needs to manage its fleet of vehicles, including cars, bikes, and trucks, and keep track of their availability. Develop a hierarchy of classes starting with a base class Vehicle and derived classes Car, Bike, and Truck. Implement methods to check

	availability, rent a vehicle, and return a vehicle.
5	A warehouse needs a system to track inventory levels, including adding new items, updating stock, and generating reports. Develop classes Item, Inventory, and Warehouse. Implement methods to add items to inventory, update stock levels, and generate inventory reports.
6	A hotel requires a booking system to manage room reservations, including checking room availability and booking rooms. Design classes Room, Hotel, and Reservation. Implement functionalities to check room availability, book a room, and display booking details.

SUBJECT NAME: Industrial Internship

SUBJECT CODE: BCA24-INT-581

CREDIT: 4

CONTACT HOURS: 48

COURSE OBJECTIVE:

1. **To provide practical exposure to the professional working environment in industry or organizations.**
2. **To enable students to apply theoretical knowledge gained during their academic program to real-world problems.**
3. **To develop technical, communication, and teamwork skills required in a professional workplace.**
4. **To familiarize students with industry practices, tools, and technologies relevant to their field of study.**
5. **To encourage professional ethics, responsibility, and problem-solving abilities through hands-on experience.**

Course Outcomes

CO1: Understand the structure, workflow, and operational practices of an industrial or organizational environment.

CO2: Apply academic knowledge and technical skills to solve practical problems during the internship.

CO3: Demonstrate professional skills such as teamwork, communication, and time management.

CO4: Prepare a technical report documenting the internship experience, tasks performed, and learning outcomes.

CO5: Present and evaluate the knowledge and skills gained during the internship through a seminar or presentation.

Internship to be started after exam of 4th sem (sem break) and completed within 5th sem (weekends)